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AMD Developer Hackathon · ACT III

WORKSIGHT

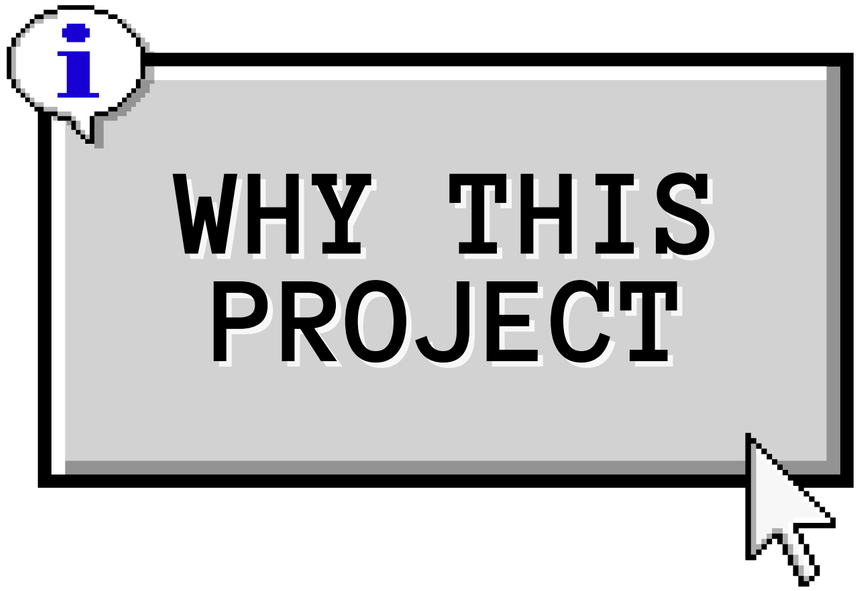
DEFECTS FROM GOOD PHOTOS



Start

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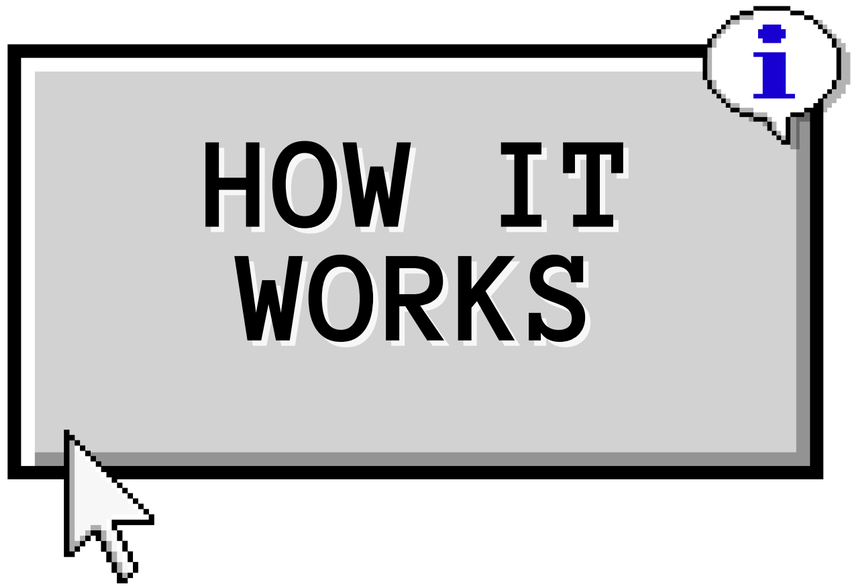
WHY THIS PROJECT

A new part on the line has no defect dataset. Defects are rare, so collecting and labelling hundreds of them takes weeks, and every new part or camera view starts again. Worksight starts from what every line already has: photos of good parts.

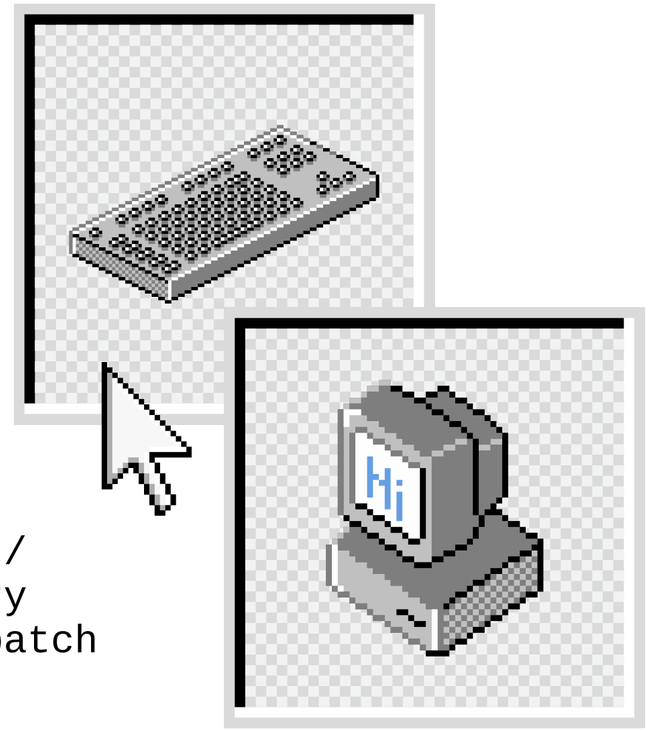
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1. 20 photos of good parts build a profile in about 9 s. No labels, no training.
2. Each part gets an anomaly score, a heatmap and the closest good reference.
3. Gemma on AMD explains what to check. The engineer decides.



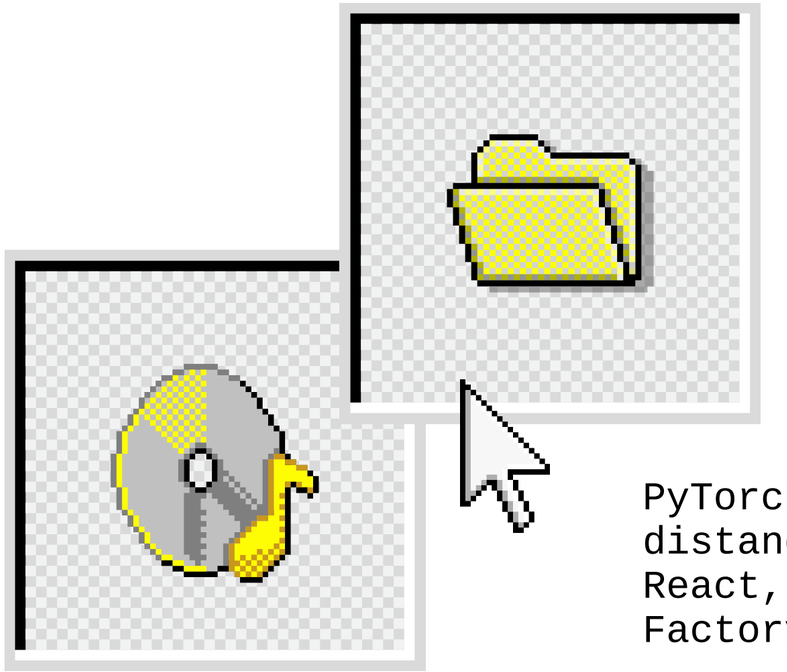
THE LIVE PRODUCT



Algorithm, AI and engineer on one screen: heatmap and closest reference, Pass / Fail / Needs review, Gemma's explanation with every inspection, one-page PDF report, history, batch inspection, 7 languages.

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TECH STACK

PyTorch: frozen ResNet-18 features, nearest-patch distance. FastAPI, SQLite, job queue. Next.js 16, React, TypeScript. Gemma-4-31B-IT on AMD Token Factory. Railway. pytest, Vitest, Playwright.

BENCHMARKS

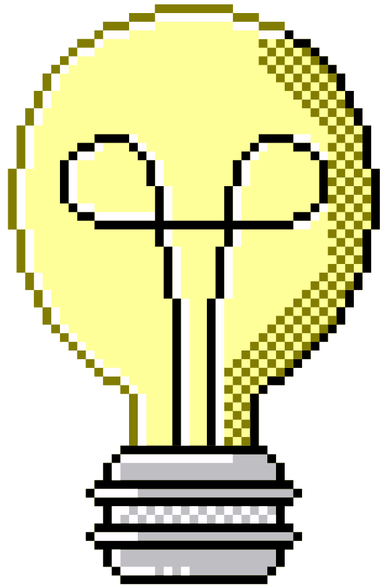
VisA PCB · held-out test · CPU

Defects caught above the threshold: pcb1 44%, pcb2 59%, pcb3 46%, pcb4 87%, with 2–6% false alarms on good boards.

With the review band, 56–96% of defects reach the engineer, who sees 4–13% of good boards. 0.19–0.30 s per image on CPU.

PARTNER TECH & WHY

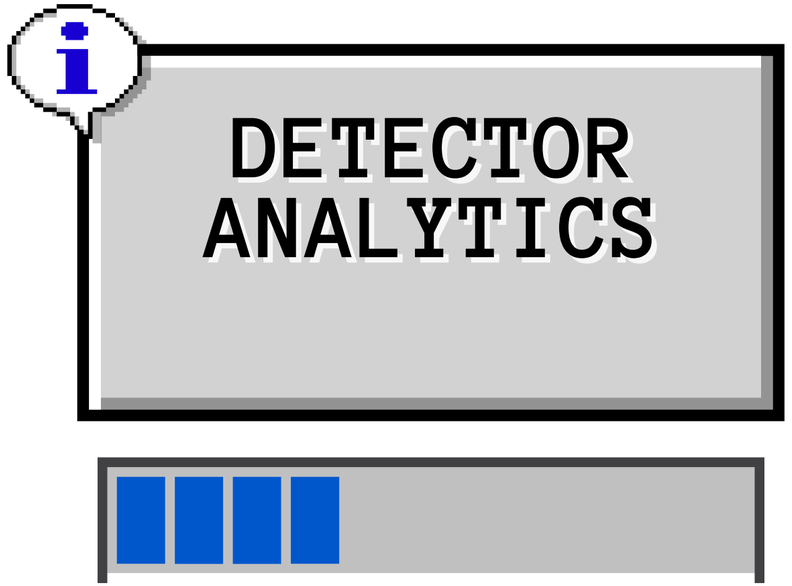
AMD Token Factory runs Gemma on AMD GPUs; the API key stays on our server. Gemma 4 31B reads the photo, heatmap, reference and part documents and explains in plain words. DeepSeek V4.1 Flash is the fallback. Next: the detector on AMD Instinct.



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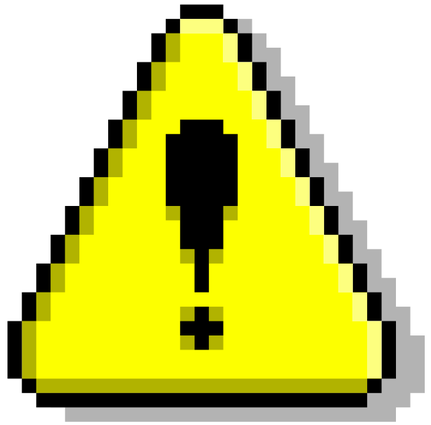
AUROC 0.88 to 0.99 on four circuit-board sets. F1 0.59 to 0.92.



v2 raised dev AUROC from 0.806 to 0.862. A 24-variant sweep found nothing better, so production stayed unchanged.

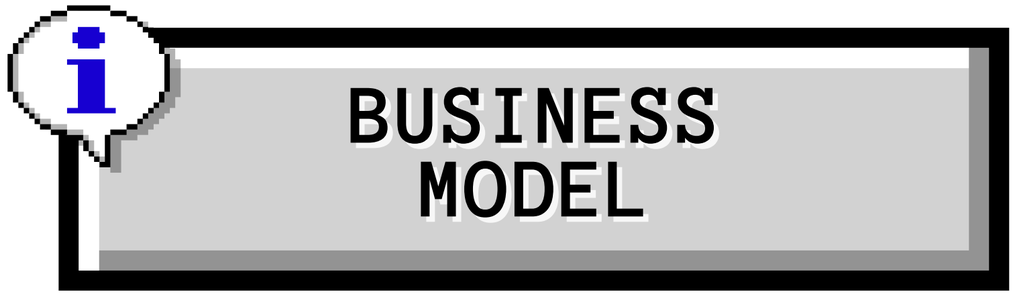
The engineer decides; model and engineer decisions are stored separately. Gemma never changes the detector result. A new profile version goes live only if it adds no false alarm or missed defect, and it can be rolled back.







AI visual inspection is forecast to grow by about $27B by 2029 (Technavio). Most tools need labelled defects or are code libraries. Worksight starts from good photos, explains each result and keeps the engineer in charge.



BUSINESS MODEL

Yes

Cancel

Subscription per camera view, or on-premises on AMD hardware for plants that keep images on site. Pilot on one line and one camera: measure engineer minutes per part and escaped defects.

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Bakhtiyar, CEO and AI engineer: detector, benchmarks, Gemma integration. Ali, frontend developer: operator workspace and reports. Next: detector on AMD Instinct, team workspaces, edge station on AMD Ryzen AI.



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